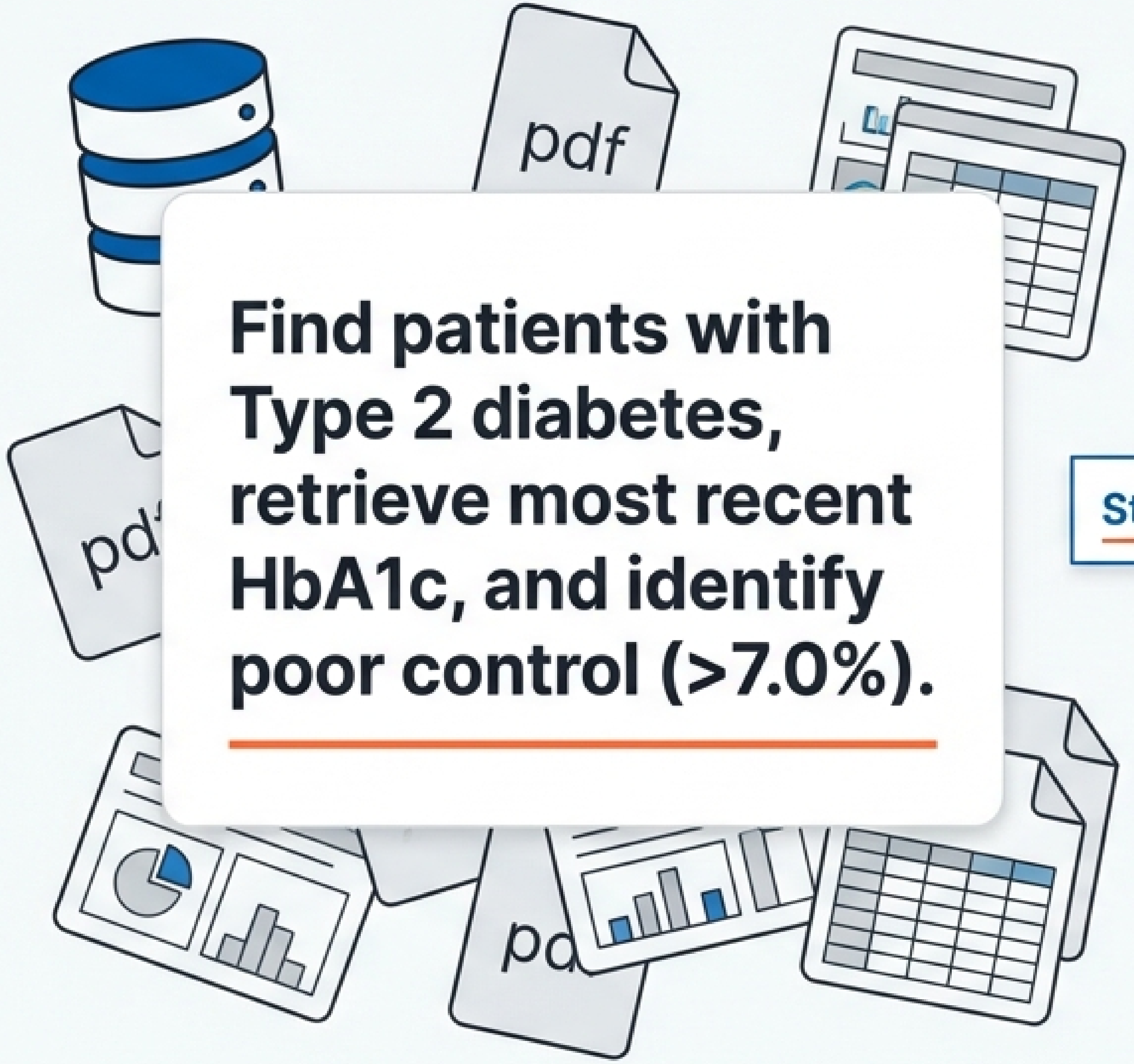
FHIR + AI Agents

From Manual Queries to Autonomous Clinical Intelligence

BASED ON THE BMI 512 CLINICAL INFORMATICS HACKATHON

Fragmented Reality



Find patients with Type 2 diabetes, retrieve most recent HbA1c, and identify poor control (>7.0%).

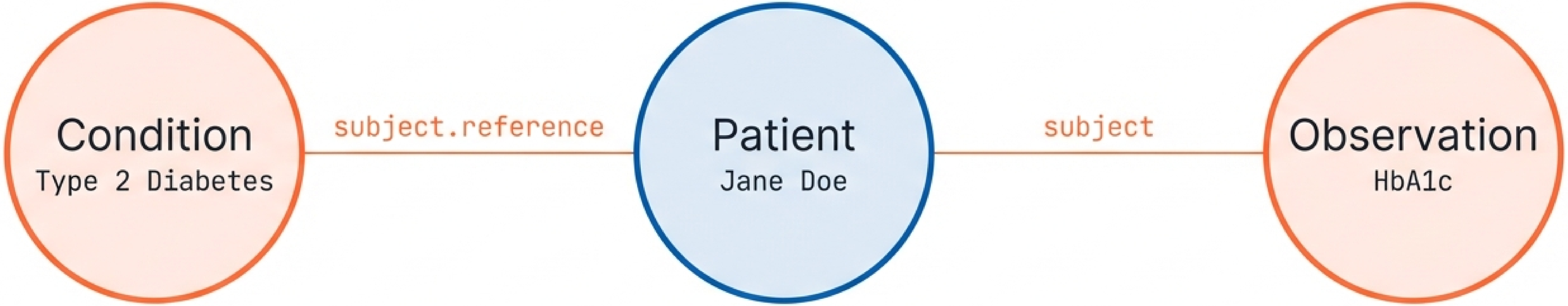
Structured Clarity

FHIR Resource


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "tetal": 1,
  "entry": [
    {
      "resource": {
        "resourceType": "Patient",
        "ld": "p-12345",
        "conditlon": [
          {
            "code": {
              "coding": [
                {
                  "system": "http://enowed.info/sct",
                  "code": "44054006",
                  "dlsptay": "Type 2 diabetes mellitus"
                }
            } ]
          ],
          "observation": [
            {
              "resourceType": "Observation",
              "2d": "obs-56789",
              "code": {
                "ceding": [
                  "system": "http://loinc.org",
                  "code": "4548-8",
                  "alsplay": "Hemoglobin A1c/Hemoglobin.total in Blood"
                ]
              },
              "valueQuantity": {
                "value": 3.5,
                "unit": "%",
                "syetem": "http://unitsofmeasure.org",
                "code": "%"
              },
              "effectiveDateTime": "2024-01-15"
            }
          }
        ]
    }
}
```


Healthcare data is fragmented. We need a standard way to ask complex questions.

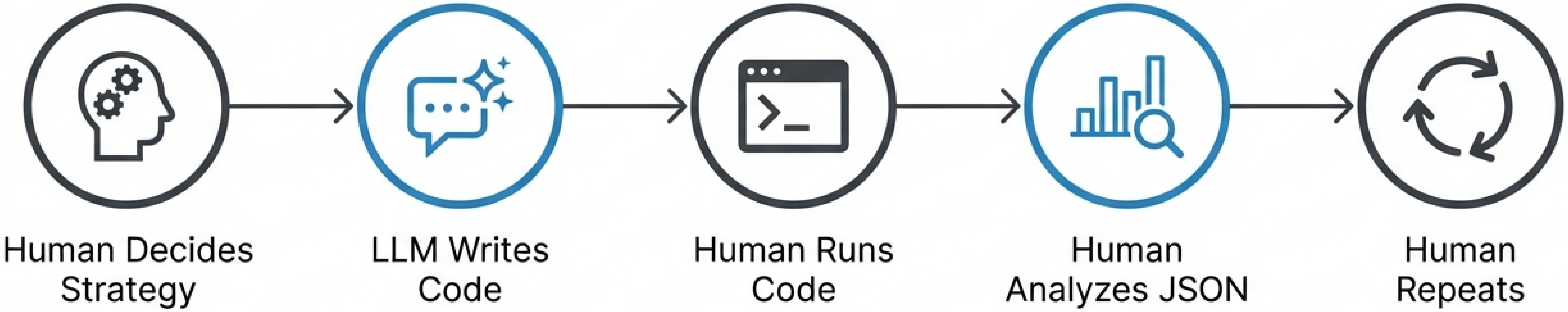
FHIR is the API for Clinical Data



Resources: The Nouns (Patient, Condition, Observation)

References: The Links (Traversing the graph)

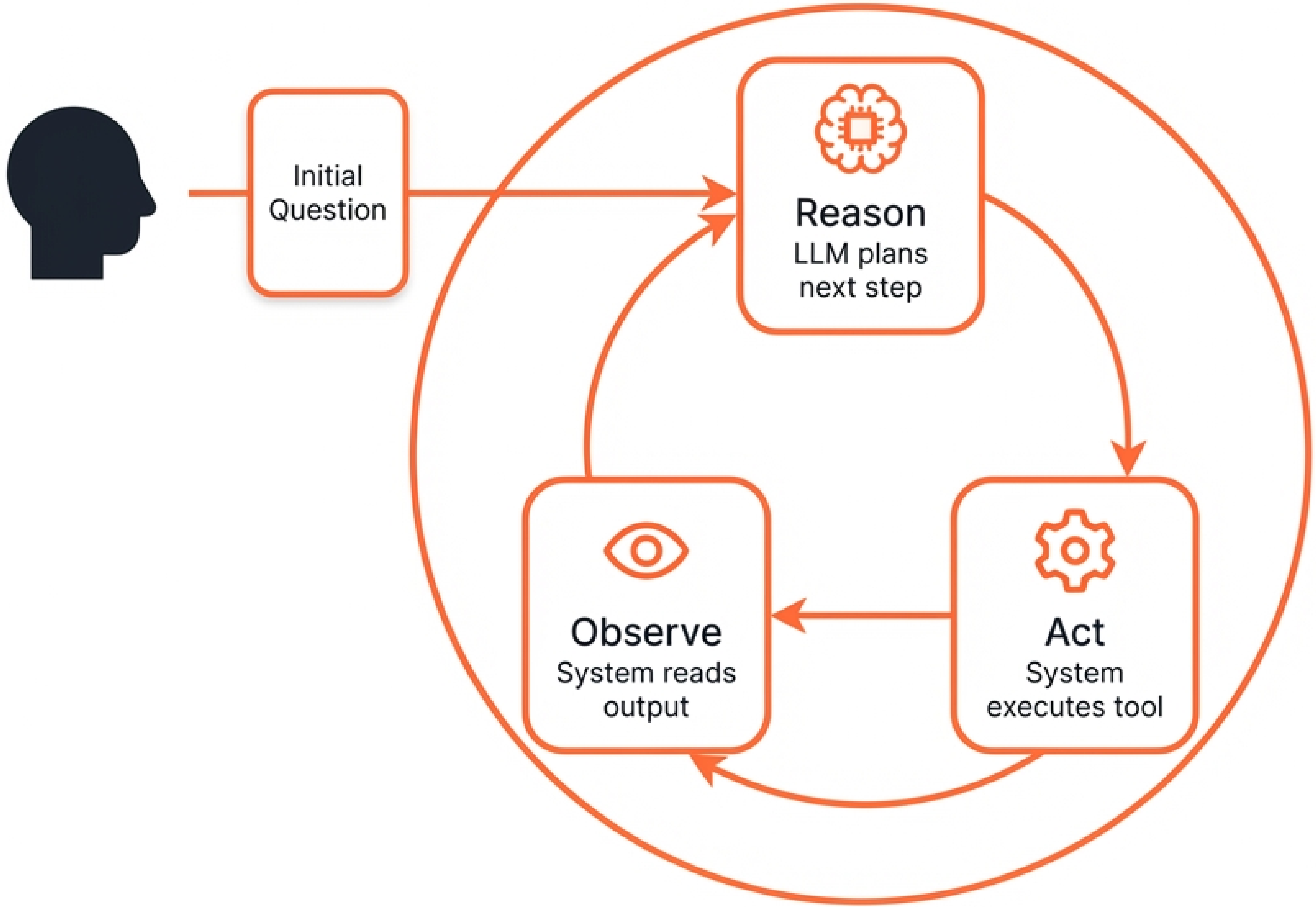
Workflow A: Manual Orchestration (Session 1)



Key Insight

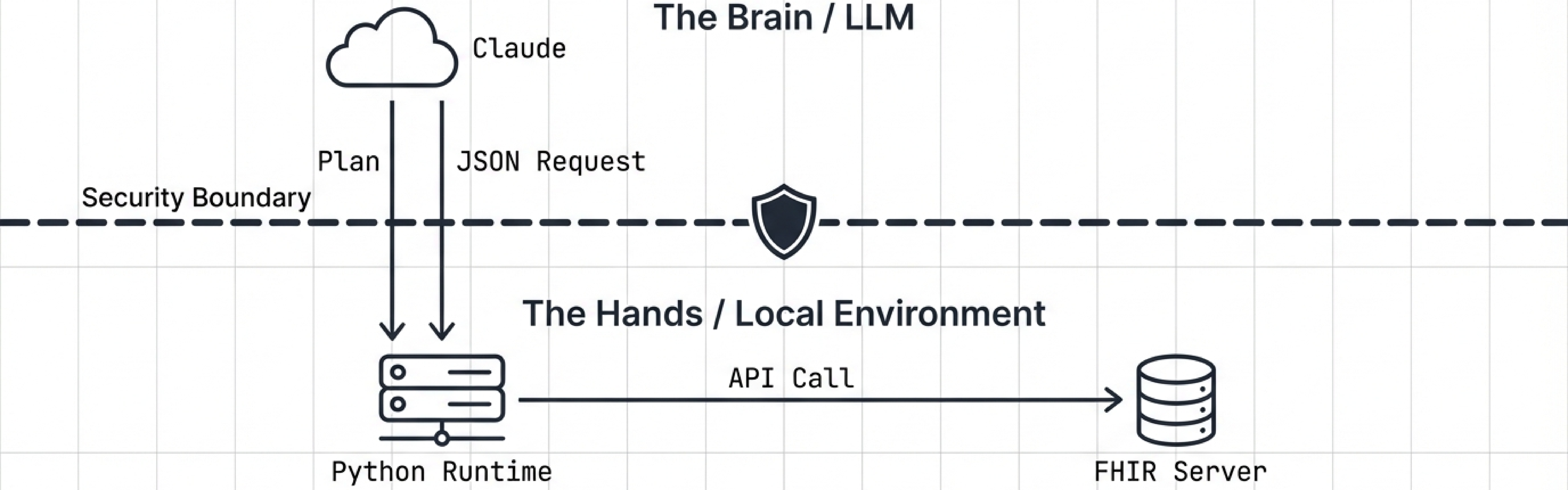
The Human is the Orchestrator. The LLM is only a coding assistant.

Workflow B: Agentic Orchestration (Session 2)




```
Tool Use (Function Calling)
allows the LLM to interact
with the external world.
```

Anatomy of an Agent



Key Insight

The LLM never touches the FHIR server directly. `YOUR CODE` does the querying.

The Instruction Manual: Tool Schemas

```
{
  "name": "search_observations",
  "description": "Find lab
  results for a patient. Common
  LOINC codes: '4548-4' for
  HbA1c.",
  "parameters": { ... }
}
```

The LLM doesn't see your code. It reads this schema to understand its capabilities.

The Standing Orders: System Prompt

Role

You are a clinical data assistant.

Constraint

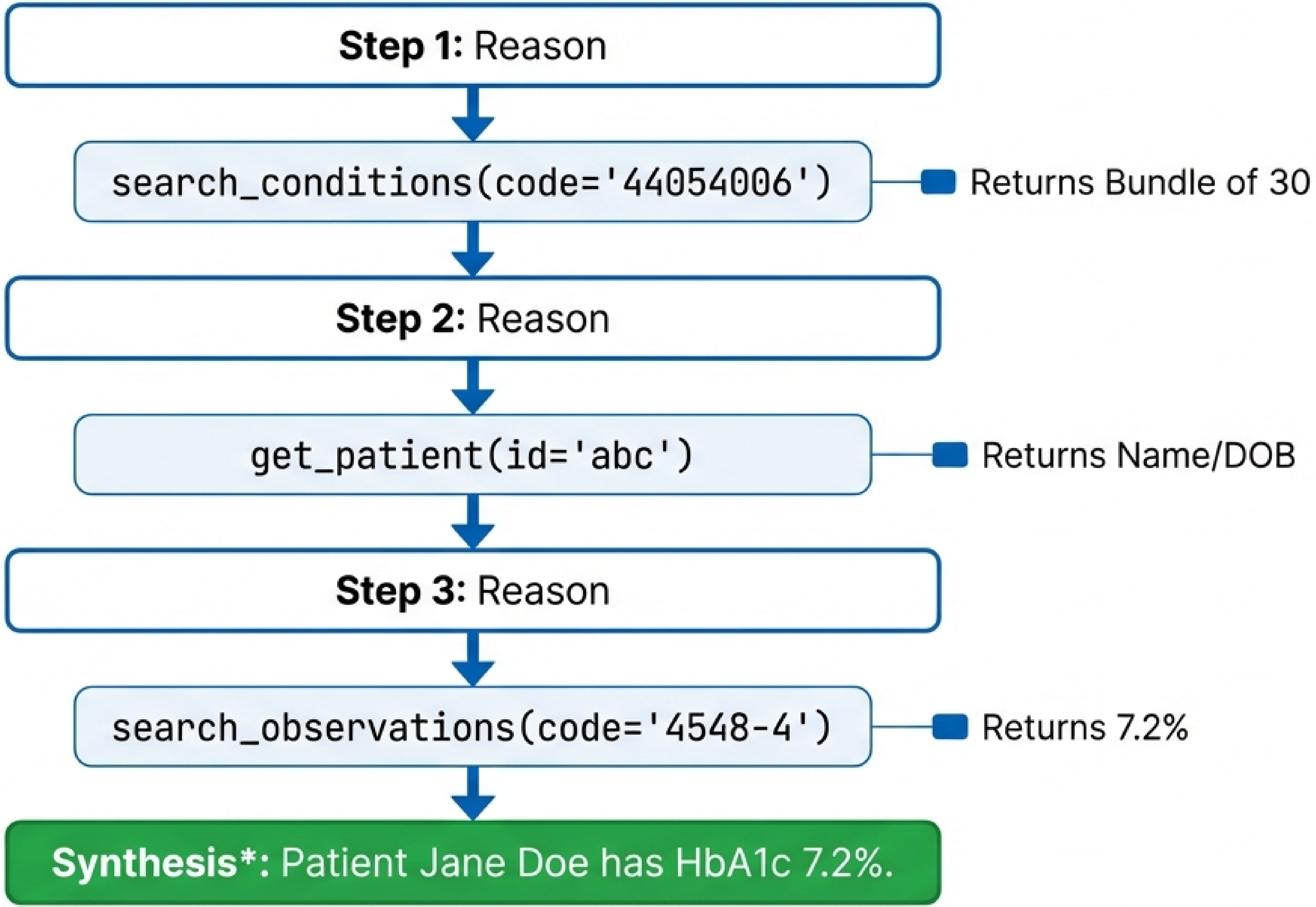
NEVER invent data.

Strategy

Think step-by-step.

`System Prompt acts as the protocol for the AI.` → ✅ **Safe Output**

The Agent Loop in Action



Translating Clinical Intent to Code

Translation Pipeline

Type 2 Diabetes $\xrightarrow{\text{SNOMED CT}}$ 44054006

HbA1c $\xrightarrow{\text{LOINC}}$ 4548-4

Poor Control $\xrightarrow{\text{Logic}}$ $> 7.0\%$

SNOMED is the dictionary of diseases. LOINC is the catalog of tests.

When Agents Fail

Hallucination

Inventing codes that don't exist in the standard.

Wrong Tool

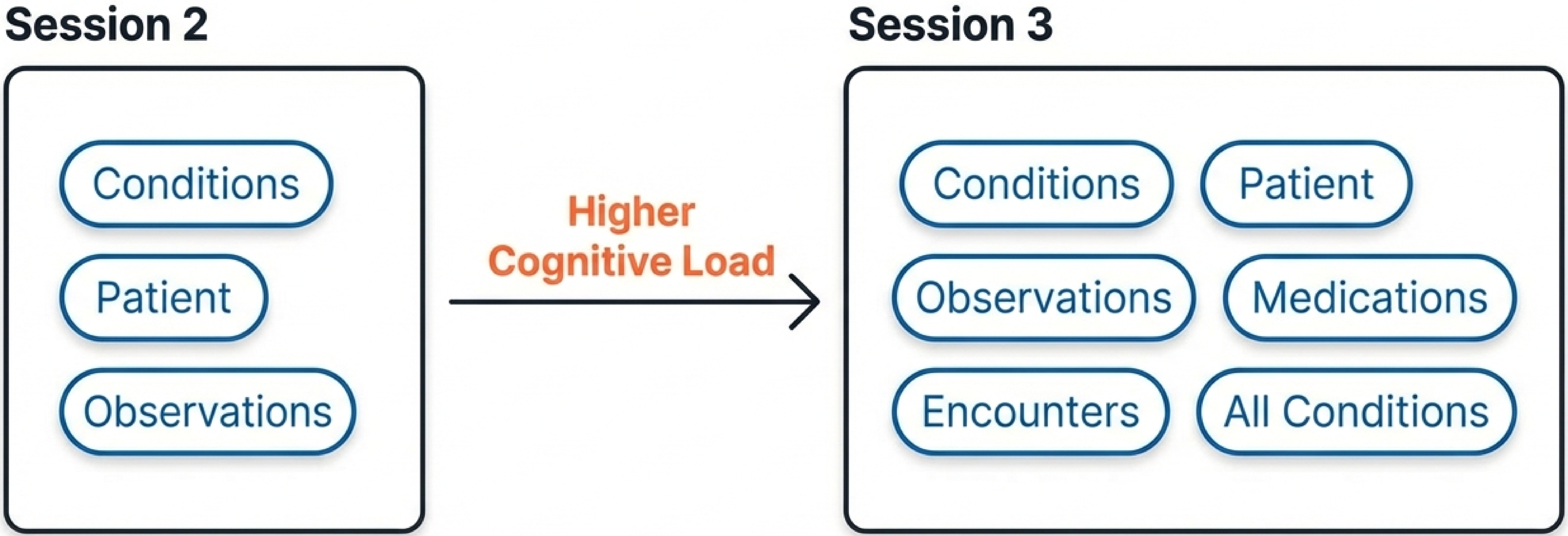
Using `search_conditions` (`Diagnosis`) instead of `search_observations` (`Lab`).

Looping

Getting stuck fetching patients one by one until step limit.

Evaluation Goal: Find these failures before deployment.

Scaling Complexity



More tools require stricter prompts to prevent the agent from getting lost in the data.

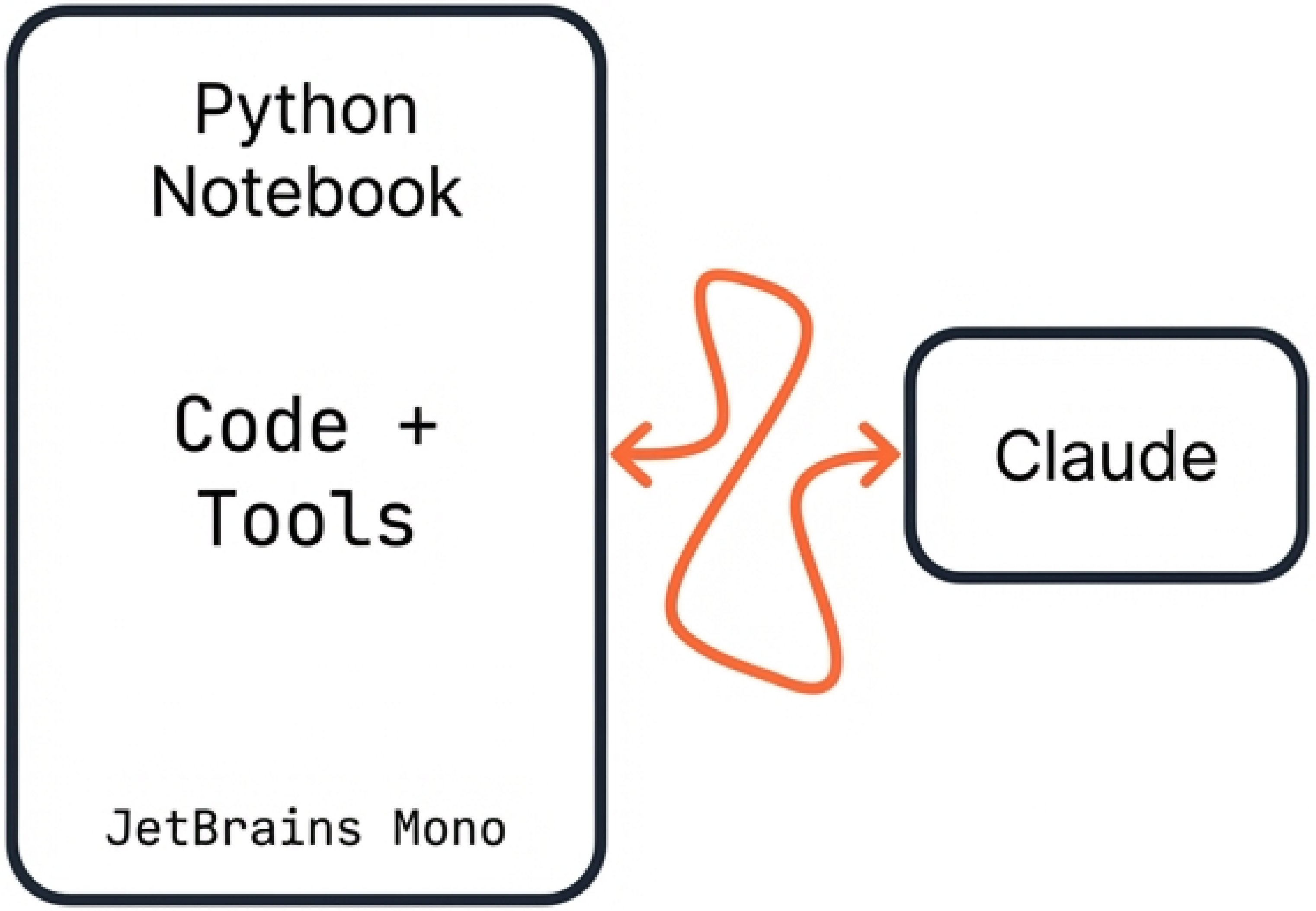
Building Trustworthiness

- **Audit Trails**
  Every tool call must be logged and visible.
- **Groundedness**
  Answers must cite specific FHIR resources.
- **Human-in-the-Loop**
  The agent assists, the clinician decides.

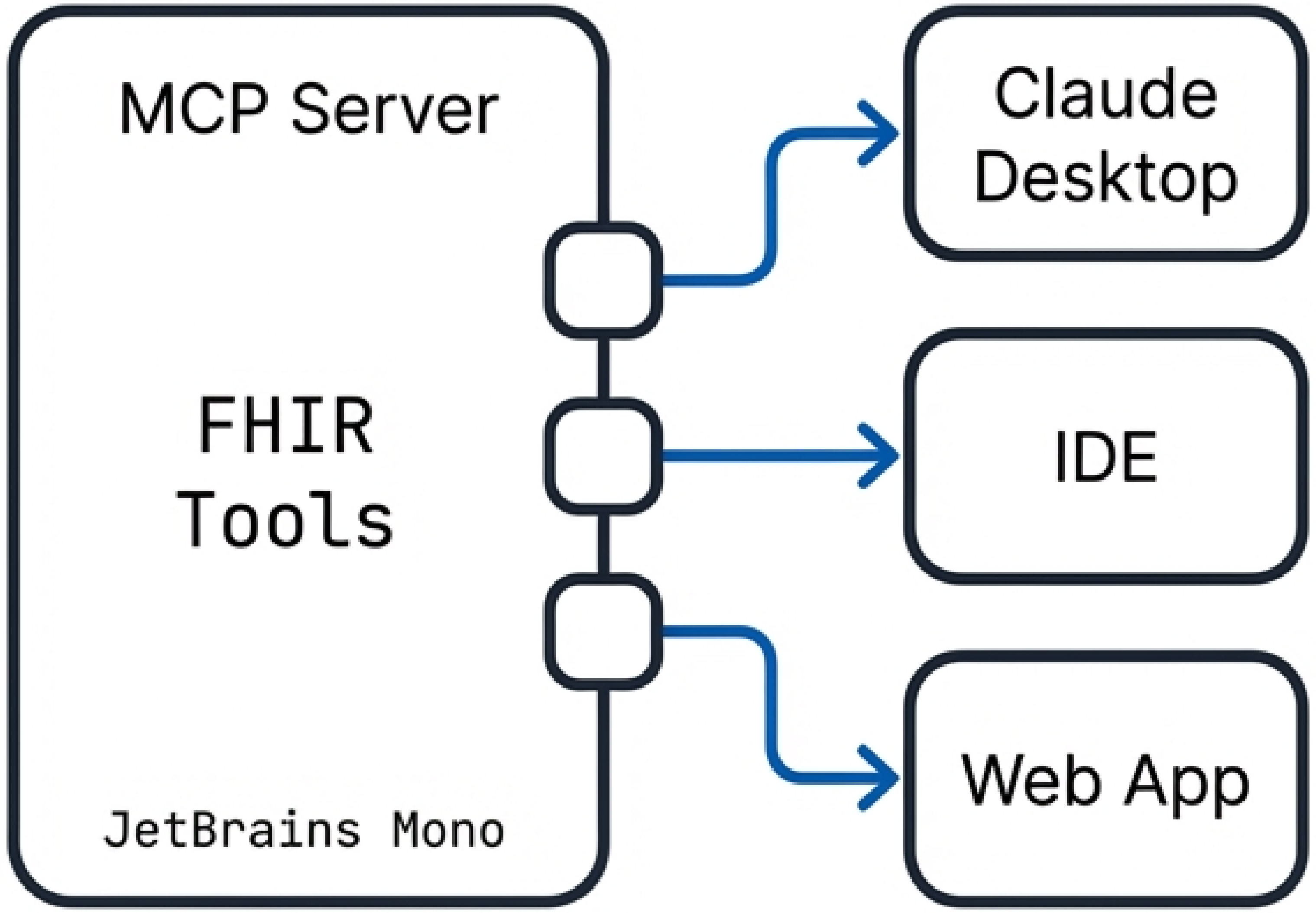
"The agent is the retrieval engine, not the doctor."

The Future: Model Context Protocol (MCP)

Notebook (Siloed)



MCP (Universal)



Standardizing discovery and connection. Build the tools once, use them with any agent.

The Evolution of Clinical Intelligence

Structure
FHIR Graph

Orchestration
Agents Plan & Act

The LLM plans, but your code executes.

Control
Schemas & Prompts

Verification
Human Trust